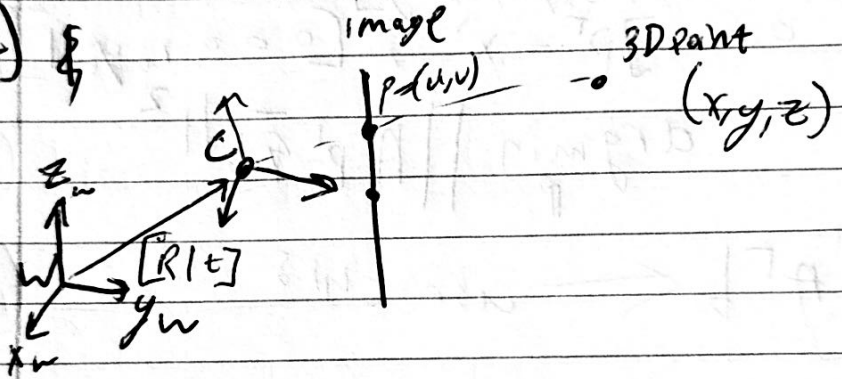
HW #8

CM-Q1) matrix form of pin-hole camera model

$$K = \begin{bmatrix} \alpha_x & s & u_0 \\ 0 & \alpha_y & v_0 \\ 0 & 0 & 1 \end{bmatrix}$$ skew

$$y' = \underbrace{K \left[R_w^c \mid t_{cw}^w \right]}_{P} X'$$

CM-Q2) &



CM-Q3) Euclidean xform - $\begin{bmatrix} c\theta & -s\theta & t_x \\ s\theta & c\theta & t_y \\ 0 & 0 & 1 \end{bmatrix}$ - 3 DOF invariant for lengths, angles, areas parallelism

Similarity - $\begin{bmatrix} s\cdot c\theta & s(-s\theta) & t_x \\ s s\theta & s c\theta & t_y \\ 0 & 0 & 1 \end{bmatrix}$ - 4 DOF - invar for angles, ~~ratios~~, parallelism

Affine - $\begin{bmatrix} A & t_x \\ & t_y \\ 0 & 1 \end{bmatrix}$ - 6 DOF - invar for parallelism, + ratios

Projective - 3x3 Hess. - 8 DOF -

LLS-Q1) affine

$x' = ax + by + t_x$

$y' = cx + dy + t_y$

$$\underset{a,b}{\operatorname{argmin}} \sum_i \|(Ax_i - b) - y_i\|^2$$

$$A = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$$

$$\underset{\begin{bmatrix} x \\ y \end{bmatrix}}{\operatorname{argmin}} \left\| A \begin{bmatrix} x \\ y \end{bmatrix} + \begin{bmatrix} t_x \\ t_y \end{bmatrix} \right\|^2$$

unknowns - $p = [a, b, t_x, c, d, t_y]$ X

$[x_i \; y_i \; 1 \; 0 \; 0 \; 0]\, p^T = x'$; $[0 \; 0 \; 0 \; x_i \; y_i \; 1]\, p^T = y'$

$Ap^T = b$ $\quad \underset{p}{\operatorname{argmin}} \|Ap^T - b\|^2$

LLS-Q2) $p = (A^T A)^{-1} A^T b$ ← use this

LLS-Q3) 3 points needed (s.t.→) & AA^T is invertible